Year 8 Science – Booklet 3: Chemistry

Periodic Table (Atoms and Elements) — ANSWERS

Quick Test (p.57)

1. It contains only one type of atom (all atoms of an element are identical, and different from atoms of any other element).
2. In order of increasing atomic number, with elements of similar properties placed in the same column (group).
3. Periods.
4. Groups.
5. The nucleus.
6. Protons and neutrons.
7. Electrons.
8. A negative charge.

Section A – Multiple Choice (5 marks)

1. b) increasing atomic number
2. a) 100
3. a) groups
4. d) electrons
5. a) protons and neutrons

Section B (15 marks)

1. True or false (10 marks)

a) Elements contain only one type of atom – **True**
b) Atoms of gold are the same as atoms of oxygen – **False**
c) Compounds are formed when atoms of two or more elements are mixed together – **False** (they must be chemically joined, not just mixed)
d) Compounds are formed when atoms of two or more elements are joined together – **True**
e) The horizontal rows in the periodic table are called periods – **True**
f) The vertical columns in the periodic table are called groups – **True**
g) The element carbon can be represented by the symbol C – **True**
h) The element chromium can be represented by the symbol C – **False** (chromium's symbol is Cr – C is carbon)
i) Protons have a negative charge – **False** (protons are positive)
j) Neutrons and protons are found in the nucleus of a carbon atom – **True**

2. Anagrams and definitions (5 marks)

moat → **atom** — a small piece of matter which was previously thought to be the smallest part of any substance
unclesu → **nucleus** — the central part of an atom which comprises neutrons and protons
leertocn → **electron** — a very small particle with a negative charge
unroten → **neutron** — a very small particle with no charge
potnor → **proton** — a very small particle with a positive charge

Section C (9 marks)

1. Copper/iron/nitrogen/oxygen information box

a). Copper (or iron).

b). Nitrogen (or oxygen).

c). Iron.

d). Copper oxide.

2. Oxygen, O■, and three particle diagrams a, b, c

a). 2.

The colours in diagrams a/b/c are hard to distinguish precisely on this scan, so this reading follows the same visual convention used elsewhere in this booklet series (single colour = element, two different colours joined = compound) – check against your own copy:

b). c – the only diagram showing two different types (colours) of atom joined together, which is what makes a compound.

c). b – paired (joined) circles of a single colour, spread apart with gaps between pairs, representing O■ gas molecules (diagram a is a tightly-packed solid of single un-paired atoms, not a gas).

d). a and b (both are made of only one colour/type of atom).

e). Elements.

Note: This is a proprietary Aristeia Academy Tuition worksheet, not an exam-board past paper, so no publicly published markscheme exists for it. These answers were composed directly from the teaching content on the preceding pages of the booklet.